

Akshay Yadav

Cybersecurity | Application Security | Red Team Assessment | Secure Software | AI for Security
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TECHNICAL SKILLS

Cybersecurity: Web Application Security Testing, Vulnerability Assessment, Red Team Assessments, Threat Modeling, OWASP Top 10, CVSS Scoring, EM Side-Channel Analysis, Digital Forensics, Networking

Secure Development & DevSecOps: Secure Coding, Input Validation, Authentication & Access Control, Unit Testing, CI/CD, Docker

AI & Automation: Google Gemini API (LLM), Prompt Engineering, Tool Calling, Workflow Automation

Programming Languages: Python, JavaScript, Java, SQL

Frameworks & Libraries: FastAPI, Flask, Pydantic, React, Node.js, Express, Next.js, Tailwind CSS

Databases: PostgreSQL, MySQL, Supabase, Redis, Firebase

Tools & Platforms: Burp Suite, Nmap, Linux, Windows, Git, GitHub, Vercel, Postman, VS Code

EXPERIENCE

Red Team Security Intern

Feb 2025 – Mar 2025

Red Team Users

Remote

- Performed threat modeling, web application security testing and vulnerability assessments on 15+ web endpoints and network assets using Burp Suite and Nmap, finding Medium to Critical issues (CVSS 6.0–8.5+)
- Documented findings with CVSS severity and clear remediation steps in a security report for the internal team

Technologies: *Burp Suite, Nmap, Linux*

PROJECTS

KeyWhisper (EM-SEC) | EM Side-Channel Attack Research Platform

2025

Contributor | github.com/AkshayByte/keywhisper-ai

- Contributed to a proof-of-concept cybersecurity platform demonstrating non-contact EM side-channel attacks that reconstruct keystrokes from keyboard electromagnetic emissions captured with an SDR
- Analyzed RF keystroke leakage across 100 frequency bins at 2.4 GHz, with neural-network keystroke classification (LSTM, CNN, Transformer) and confidence-scored reports
- Hardened Flask API with upload sanitization and route isolation, eliminating unhandled 400/500 crashes
- Built an async UI toast engine, preventing render-loop freezes during 60 FPS Canvas spectrogram runs

Technologies: *Python, Flask, JavaScript, HTML5 Canvas, REST API*

SwasthaTrack | AI-Assisted Healthcare Platform

Jan 2025 – Present

Developer | github.com/AkshayByte/swasthatrack | swasthatrack.vercel.app

- Built an asynchronous FastAPI triage pipeline using the Google Gemini API (LLM) with prompt engineering and tool calling, automating priority queues with 90%+ diagnostic accuracy
- Engineered 15+ secure REST endpoints with Pydantic input validation, authentication and role-based access control, syncing sensitive clinical data across 5 dashboards with sub-150ms state latency
- Wrote unit tests and automated CI/CD with GitHub Actions; containerized with Docker and deployed on Vercel

Technologies: *JavaScript, FastAPI, PostgreSQL, Gemini API, Docker, GitHub Actions, Tailwind CSS*

EDUCATION

VIT Bhopal University

2023 – 2027

B.Tech, Cybersecurity & Digital Forensics (Final Year) | CGPA: 8.72/10.0

Bhopal, India

CERTIFICATIONS

- The Bits and Bytes of Computer Networking — Google
- Blockchain and its Applications — IIT Kharagpur (NPTEL)